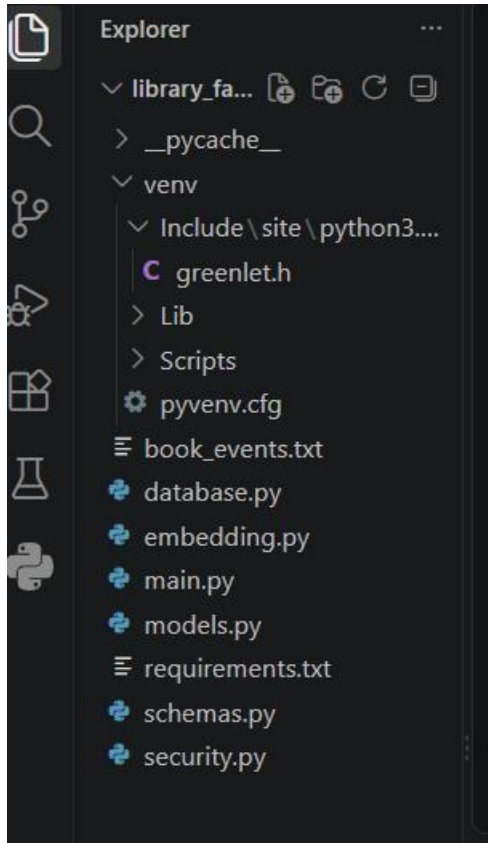


PRACTICAL
WEEK-5
2520090046

1. Project Structure

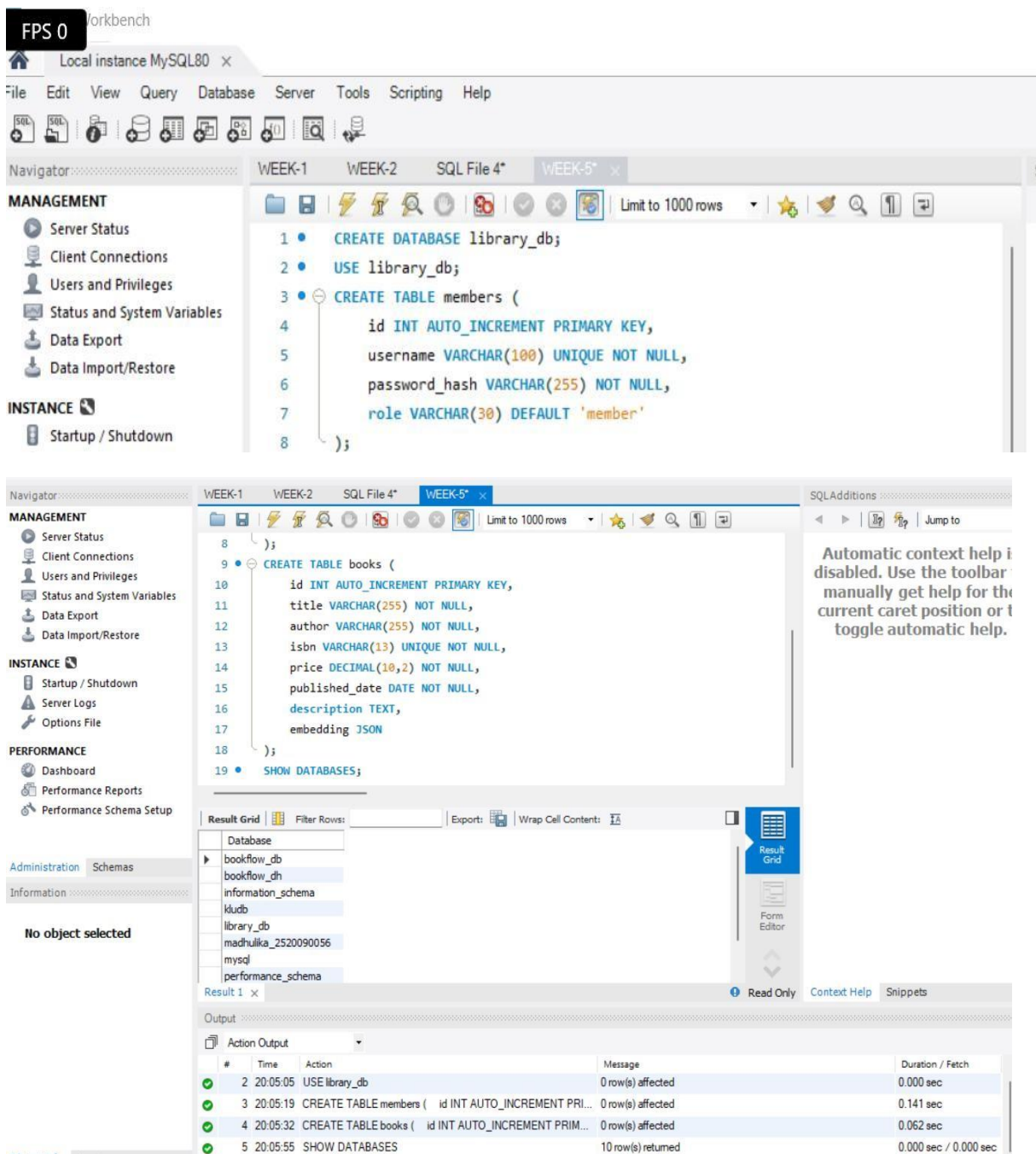


2. Installed Packages

```
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 6375, in startup
    await handler()
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\main.py", line 38, in startup
    async with engine.begin() as connection:
File "C:\Users\shivani\AppData\Local\Programs\Python\Python311\Lib\contextlib.py", line 210, in __aenter__
    return await anext(self.gen)
    ~~~~~
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\engine.py", line 1068, in begin
    async with conn:
```

```
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\base.py", line 122, in __aenter__
    return await self.start(is_ctxmanager=True)
    ~~~~~
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 144, in __init__
    self.dispatch.connect(self)
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 3319, in raw_connection
    return self.pool.connect()
    ~~~~~
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 448, in connect
    return _connectionfairy.checkout(self)
    ~~~~~
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _checkout
    fairy = _connectionRecord.checkout(pool)
    ~~~~~
File "C:\Users\shivani\OneDrive\Desktop\DBSS\Library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in _checkout
    rec = pool._do_get()
    ~~~~~
```

3. MySQL Setup



Navigator: WEEK-1 WEEK-2 SQL File 4* WEEK-5*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

10 id INT AUTO_INCREMENT PRIMARY KEY,
11 title VARCHAR(255) NOT NULL,
12 author VARCHAR(255) NOT NULL,
13 isbn VARCHAR(13) UNIQUE NOT NULL,
14 price DECIMAL(10,2) NOT NULL,
15 published_date DATE NOT NULL,
16 description TEXT,
17 embedding JSON
18 );
19 SHOW DATABASES;
20 USE library_db;
21 SHOW TABLES;

```

Result Grid: Filter Rows: Exports: Wrap Cell Content: **Result Grid**

Tables_in_library_db

- books
- members

Result 2 x Read Only

Output

Action Output

#	Time	Action	Message
4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected
5	20:05:55	SHOW DATABASES	10 row(s) returned
6	20:07:46	USE library_db	0 row(s) affected
7	20:07:52	SHOW TABLES	2 row(s) returned

Object Info Session

4. Database.py

Explorer: library_fastapi_project

- __pycache__
- venv
- Include\site\python3...
- greenlet.h
- Lib
- Scripts
- pyvenv.cfg
- book_events.txt
- database.py
- embedding.py
- main.py
- models.py
- requirements.txt
- schemas.py
- security.py

database.py

```

1 from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker, create_async_engine
2 from sqlalchemy.orm import DeclarativeBase
3
4 DATABASE_URL = (
5     "mysql+asyncmy://root:Madhu%40248"
6     "@localhost:3306/library_db"
7 )
8
9 engine = create_async_engine(
10     DATABASE_URL,
11     echo=True
12 )
13
14 AsyncSessionLocal = async_sessionmaker(
15     engine,
16     expire_on_commit=False
17 )
18
19
20 class Base(DeclarativeBase):
21     pass
22
23
24 async def get_db():
25     async with AsyncSessionLocal() as session:
26         yield session

```

Outline

5. Embedding.py

```
library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3...
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── ...

embedding.py > text_to_vector
1  import hashlib
2  import math
3
4
5  VECTOR_DIM = 384
6
7
8  def text_to_vector(text: str):
9      vector = []
10
11      for i in range(VECTOR_DIM):
12          h = hashlib.sha256(
13              f"{text}-{i}".encode()
14          ).digest()
15
16          value = int.from_bytes(h[:4], "big") / (2**32 - 1)
17          vector.append(value)
18
19      norm = math.sqrt(
20          sum(x * x for x in vector)
21      )
22
23      if norm == 0:
24          return vector
25
26      return [x / norm for x in vector]
```

6. Main.py

```
library_fa...
├── __pycache__
├── venv
├── Include\site\python3...
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── ...

main.py > login
1  from fastapi import (
2      BackgroundTasks,
3      Depends,
4      FastAPI,
5      HTTPException,
6      Query,
7      status
8  )
9
10 from fastapi.security import OAuth2PasswordRequestForm
11
12 from sqlalchemy import select
13 from sqlalchemy.ext.asyncio import AsyncSession
14
15 from database import Base, engine, get_db
16 from embedding import text_to_vector
17 from models import Book, Member
18 from schemas import BookSchema, MemberRegister, TokenResponse
19
20 from security import (
21     create_access_token,
22     get_current_member,
23     hash_password,
24     verify_password
25 )
26
27 import math
28
29
30 app = FastAPI(
31     title="Library Backend & Security API",
32     version="1.0"
33 )
34
35
36 @app.on_event("startup")
```

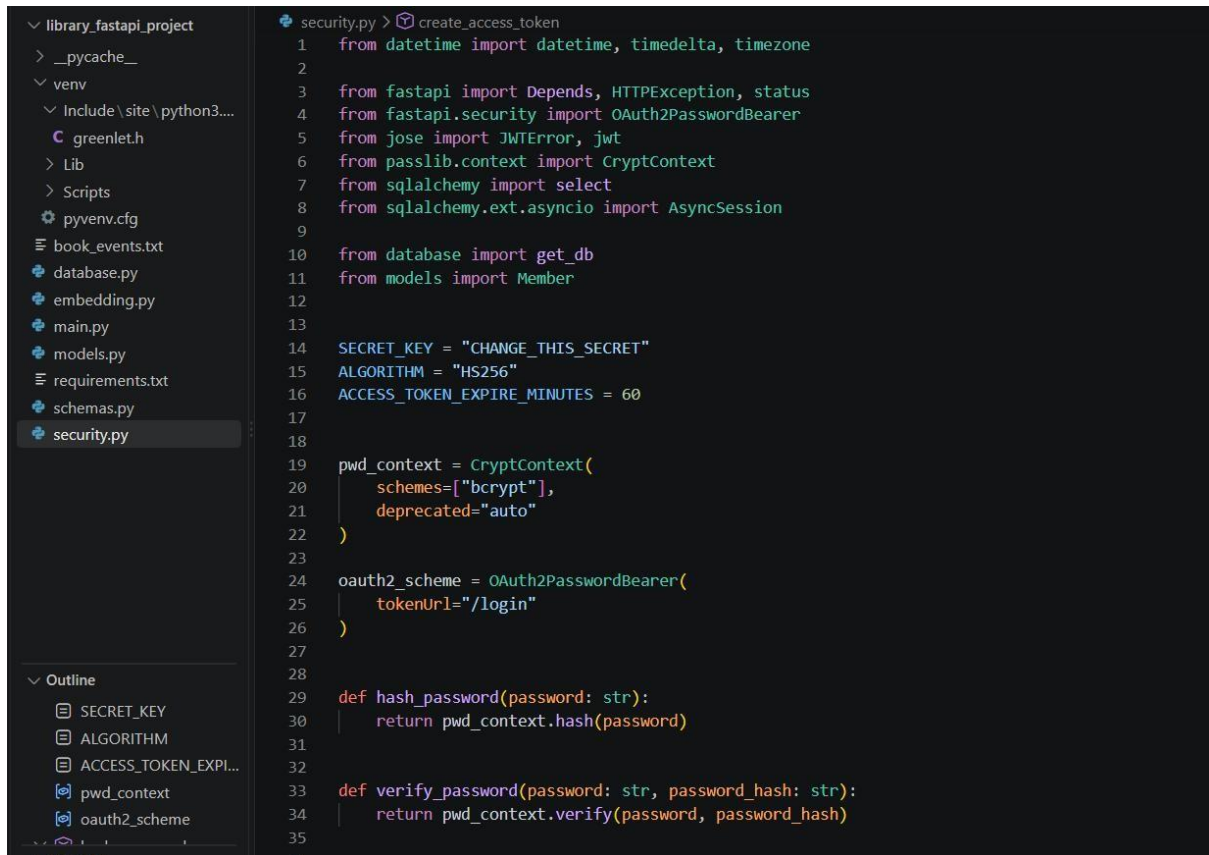
7. Models.py

```
library_fastapi_project
├── __pycache__
├── venv
│   └── Include\site\python3....
│       ├── greenlet.h
│       ├── Lib
│       ├── Scripts
│       ├── pyvenv.cfg
│       ├── book_events.txt
│       ├── database.py
│       ├── embedding.py
│       ├── main.py
│       ├── models.py
│       ├── requirements.txt
│       ├── schemas.py
│       └── security.py
└── models.py > Book
1  from sqlalchemy import Column, Integer, String, Text, DECIMAL, Date, JSON
2  from database import Base
3
4
5  class Member(Base):
6      __tablename__ = "members"
7
8      id = Column(Integer, primary_key=True, index=True)
9      username = Column(String(100), unique=True, nullable=False)
10     password_hash = Column(String(255), nullable=False)
11     role = Column(String(30), default="member")
12
13
14 class Book(Base):
15     __tablename__ = "books"
16
17     id = Column(Integer, primary_key=True, index=True)
18     title = Column(String(255), nullable=False)
19     author = Column(String(255), nullable=False)
20     isbn = Column(String(13), unique=True, nullable=False)
21     price = Column(DECIMAL(10, 2), nullable=False)
22     published_date = Column(Date, nullable=False)
23     description = Column(Text)
24     embedding = Column(JSON)
```

8. Schemas.py

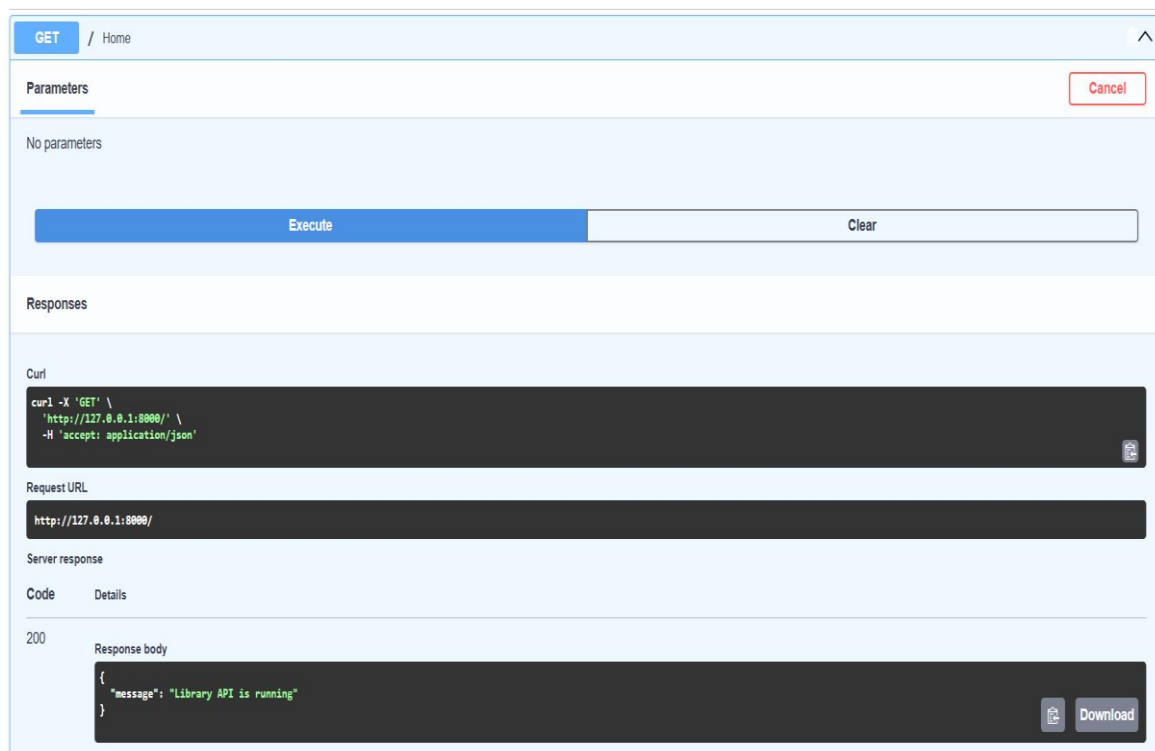
```
Explorer
├── library_fastapi_project
│   ├── __pycache__
│   ├── venv
│   │   └── Include\site\python3....
│   │       ├── greenlet.h
│   │       ├── Lib
│   │       ├── Scripts
│   │       ├── pyvenv.cfg
│   │       ├── book_events.txt
│   │       ├── database.py
│   │       ├── embedding.py
│   │       ├── main.py
│   │       ├── models.py
│   │       ├── requirements.txt
│   │       ├── schemas.py
│   │       └── security.py
│   └── schemas.py
└── schemas.py > MemberRegister
1  from pydantic import BaseModel, Field, field_validator
2  from datetime import date
3
4
5  class BookSchema(BaseModel):
6      title: str
7      author: str
8      isbn: str = Field(min_length=13, max_length=13)
9      price: float = Field(gt=0)
10     published_date: date
11     description: str | None = None
12
13     @field_validator("published_date")
14     @classmethod
15     def validate_published_date(cls, value):
16         if value > date.today():
17             raise ValueError("published_date cannot be in the future")
18         return value
19
20
21 class MemberRegister(BaseModel):
22     username: str
23     password: str = Field(min_length=6)
24
25
26 class TokenResponse(BaseModel):
27     access_token: str
28     token_type: str
```

9. Security.py



```
security.py > create_access_token
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18
19 pwd_context = CryptContext(
20     schemes=["bcrypt"],
21     deprecated="auto"
22 )
23
24 oauth2_scheme = OAuth2PasswordBearer(
25     tokenUrl="/login"
26 )
27
28
29 def hash_password(password: str):
30     return pwd_context.hash(password)
31
32
33 def verify_password(password: str, password_hash: str):
34     return pwd_context.verify(password, password_hash)
35
36
```

10. Test Registration



GET / Home

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body <pre>{ "message": "Library API is running" }</pre>

Download

POST /register Register

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value Schema

```
{
  "username": "Madhu",
  "password": "library123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Madhu",
    "password": "library123"
  }'
```

Request URL

http://127.0.0.1:8080/register

Server response

Code

Details

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "Madhu",
  "role": "member"
}
```

Download

Response headers

```
content-length: 79
content-type: application/json
date: Mon, 14 Sep 2026 16:05:16 GMT
server: uvicorn
```

Responses

11. Test Login

POST

/login Login

⌵

Parameters

Cancel

Reset

No parameters

Request body required

application/x-www-form-urlencoded ▾

grant_type

string | (string | null)
pattern: "password\$"

☐ Send empty value

username * required

string

☐ Send empty value

password * required

string(\$password)

☐ Send empty value

scope

string

☒ Send empty value

client_id

string | (string | null)

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/login' \
-H 'accept: application/json' \
-H 'Content-Type: application/x-www-form-urlencoded' \
-d 'grant_type=password&username=Madhu&password=library123&scope=&client_id=Madhu&client_secret=library123'
```

Request URL

```
http://127.0.0.1:8000/login
```

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVCl9.eyJzdWI0IjNVMWModSizInJvbGU0Ij7cZWIiZXIiLCJleHAiOjE3ODQwMDU5Nzh9.BSsYLTpIyOmAoCo0hw/3PUGLxvB9I2uzBd0SyfyJVx4", "token_type": "bearer" }</pre> Response headers <pre>content-length: 186 content-type: application/json date: Mon, 14 Sep 2026 16:12:57 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json ▼

Controls Accept header.

Example Value | Schema

12. Test POST/Books

POST /books Create Book

Parameters

No parameters

Request body required

application/json

Edit Value Schema

```
{
  "title": "Python for Beginners",
  "author": "John Smith",
  "isbn": "123",
  "price": -100.00,
  "published_date": "2024-06-10",
  "description": "An introduction to Python programming."
}
```

Execute

Clear

Responses

```
curl -X POST \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVjQ9Lm5kaWkiOiJBNiIzX1IjLCJleHAiOiE3ODk4MDVhbnJ7S9.Lq7z0U6THNKUwQH8zXEgTX64tTGk1shHu43ghNn61I0' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Python for Beginners",
    "author": "John Smith",
    "isbn": "123",
    "price": 450.00,
    "published_date": "2024-06-10",
    "description": "An introduction to Python programming."
  }'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

422

Error: Unprocessable Entity

Response body

```
{
  "detail": [
    {
      "type": "string_too_short",
      "loc": [
        "body",
        "isbn"
      ],
      "msg": "String should have at least 13 characters",
      "input": "123",
      "ctx": {
        "min_length": 13
      }
    }
  ]
}
```

Download

Response headers

```
content-length: 150
content-type: application/json
date: Mon, 14 Sep 2026 16:20:41 GMT
server: uvicorn
```

GET

/books/search

Search Books

Parameters

Cancel

Name	Description
q * required	
string	q = python
(query)	minLength: 1
limit	
integer	5
(query)	maximum: 20 minimum: 1

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=q%20%3D%20python&limit=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=q%20%3D%20python&limit=5
```

Server response

13. book_event.txt

library_fastapi_project

> __pycache__

venv

> Include\site\python3...

greenlet.h

> Lib

> Scripts

pyvenv.cfg

book_events.txt

database.py

embedding.py

main.py

models.py

requirements.txt

schemas.py

security.py

book_events.txt

1 New Book Added: id=1, title=Python for Beginners

2

